Problem Set 3

AI 501: Mathematics for Artificial Intelligence | Fall 2026

Problem 1

Let

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} -1 & 2 & 0 \\ 3 & 1 & 2 \end{pmatrix}.$$

Calculate:

(a) $A + B$;

(b) $3A - 2B$;

(c) A^T and B^T;

(d) $\|A\|_F$, the Frobenius norm of A.

Problem 2

Let

$$A = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 3 & 4 \\ -2 & 0 & 5 \end{pmatrix}.$$

(a) Compute $\operatorname{tr}(A)$.

(b) Decompose $A = S + K$, where S is symmetric and K is skew-symmetric.

(c) Verify that $S^T = S$ and $K^T = -K$.

(d) Is A itself symmetric, skew-symmetric, or neither? Justify your answer using one of the parts above.

Problem 3

For each statement below, determine whether it is *always true*, *sometimes true*, or *never true*. Give a counterexample when appropriate.

(a) If A is symmetric, then A^2 is symmetric.

(b) If A and B are both symmetric $n \times n$ matrices, then AB is symmetric.

(c) The diagonal entries of a skew-symmetric matrix are all zero.

(d) $\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$.

(e) $\operatorname{tr}(AB) = \operatorname{tr}(A)\operatorname{tr}(B)$.

Problem 4

Let

$$A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix}, \qquad x = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}.$$

(a) Compute $y = Ax$ directly using the row-by-row definition of matrix-vector multiplication.

(b) Recompute y by writing it as a linear combination of the columns of A, weighted by the entries of x.

(c) Confirm that both methods give the same answer.

Problem 5

A store chain stocks $n = 4$ products across $m = 3$ warehouses. The quantity matrix is

$$Q = \begin{pmatrix} 10 & 0 & 5 & 2 \\ 4 & 6 & 0 & 8 \\ 0 & 3 & 7 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 4},$$

where Q_{ij} is the quantity of product j held at warehouse i. The unit price vector for the four products is

$$p = \begin{pmatrix} 5 \\ 10 \\ 2 \\ 7 \end{pmatrix}.$$

(a) Compute $v = Qp$, and interpret each entry of v in words.

(b) Suppose a competitor's reference stock profile is $q = (1, 1, 1, 1)^T$. Compute $Q^T w$ for some warehouse-weight vector $w = (1, 0, 0)^T$, and explain what this product represents in terms of Warehouse 1.

Problem 6

Let

$$u = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad v = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

(a) Compute the outer product uv^T.

(b) Compute the Gram matrix G of the set $\{u, v\}$, i.e., the matrix whose (i, j) entry is the inner product of the i-th and j-th vectors in the list (u, v).

(c) Is G symmetric? Explain why this must always be true for a Gram matrix.

Problem 7

Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

(a) Compute AB and BA.

(b) Are AB and BA equal? What general property of matrix multiplication does this illustrate?

(c) Compute $(AB)^T$ and $B^T A^T$ separately, and verify that they are equal.

Problem 8

A two-layer neural network computes $a^{[1]} = W^{[1]}x$ and $y = W^{[2]}a^{[1]}$, where

$$W^{[1]} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}, \qquad W^{[2]} = \begin{pmatrix} 2 & 1 \end{pmatrix}, \qquad x = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

(a) Compute $a^{[1]}$ and then y.

(b) Show that y can also be written as $y = Wx$ for a single matrix W, and find W.

(c) In one or two sentences, explain why a stack of layers with no nonlinearity between them collapses into a single linear map.

Problem 9

Consider the following systems $Ax = b$. For each, write A and b, and state whether the system is under-determined, square, or over-determined; then determine whether it has no solution, a unique solution, or infinitely many solutions.

(a) $x_1 + x_2 = 3, \quad 2x_1 + 2x_2 = 6.$

(b) $x_1 + x_2 = 1, \quad x_1 - x_2 = 3, \quad 2x_1 = 5.$

(c) $x_1 + 2x_2 - x_3 = 4, \quad -x_1 + x_2 + x_3 = 0.$

Problem 10

Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

(a) Is A tall, wide, or square? Could A possibly have a left inverse, a right inverse, both, or neither? Justify using the shape of A alone, before computing anything.

(b) Verify that

$$X = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

is a left inverse of A by checking $XA = I$.

(c) Using X, is X also a right inverse of A? Compute AX to check.

Problem 11

Let $A \in \mathbb{R}^{m \times n}$ with $m > n$ and suppose A has linearly independent columns.

(a) Explain why A cannot be right-invertible.

(b) Write down the formula for the (left) pseudo-inverse $A^\dagger$ of A in terms of A and A^T.

(c) For

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix},$$

compute $A^T A$, then $A^\dagger$.

(d) Verify that $A^\dagger A = I_2$.

Problem 12

Let $u, v, w \in \mathbb{R}^n$ be nonzero vectors and let A, B, C be matrices of compatible sizes. For each statement, determine whether it is true or false. If false, give a counterexample.

(a) If A has a left inverse, then the columns of A are linearly independent.

(b) If A is square and has a left inverse, then it also has a right inverse.

(c) A tall matrix ($m > n$) can be right-invertible.

(d) $(ABC)^T = A^T B^T C^T$.

(e) If A has orthonormal columns, then $\|Ax\| = \|x\|$ for every x.

(f) Every invertible matrix has orthonormal columns.

Problem 13

A linear classifier maps a d-dimensional feature vector x to a score using a weight matrix $W \in \mathbb{R}^{k \times d}$, one row per class:

$$z = Wx, \qquad z \in \mathbb{R}^k.$$

Suppose $d = 3$, $k = 2$, and

$$W = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \end{pmatrix}, \qquad x = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}.$$

(a) Compute $z = Wx$, and state which class score is higher.

(b) Viewing $z = Wx$ as a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$, explain in one sentence what property of the map $x \mapsto Wx$ guarantees that

$$W(x + x') = Wx + Wx', \qquad W(\alpha x) = \alpha(Wx)$$

for any vectors x, x' and scalar α.

(c) Verify the additivity property above directly using W and the two vectors $x = (2, 1, 3)^T$ and $x' = (1, 1, 0)^T$.